Introduction

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R.A Fisher → Father of statistics

Defination: According to R.A Fisher,

"The science of statistics is essentially a branch of Applied mathmatics and may be regarded as mathmatics applied to observational data."

* The theory of decision-making in the face of uncertainty.

Thus, statistics can also be defined as a branch of sciench which deals with collection, organization, presentation and analysis of numerical data and drawing valid inference therefrom.

Statistics in Agriculture:

Widely used in all the sectors (physical, biological, economic and phychology ...)

- Agricultural census are conducted to collect information like, total cultivated land, cultivated crop, fertilizer use. After collection those are organized and presented.
- Statistical techniques are applied to analysis those data on the basis of regional requirement of fertilizer and irrigation.
- We are often interested in comparing a set of varieties, agricultural practices or manurial treatments. The very important branch of statistics known as the design and analysis of experiments has widest application in agriculture.
- Thus it is often possible to identify the best variety or the best comb^{n} of variety and chemical fertilizer under given climate and soil conditions.
- Also analysis of various data on veterinary treatment and Fish culture.

Limitations of Statistics:

a) Statistics does not refer to the characteristics of any individual, rather it analyses the collected data and refers to the overall results.

b) Statistical result are true on the average, for particular case, it may not be true.

c) Statistic usually collects data through sample survey and comments on the population characteristics on the basis of sample information.

⊛ Population: An aggregate of elements possessing certain characteristics of interest in any particular investigation or enquary.
- If all the elements of the population are countable then it is called finite population, if not then it is called infinite population.

⊛ Sample: A sample is a representative part of the population.
The properties of the population is important to know. Sometime it is impractical or even impossible to handle the population. Thats why inferences about the population are usually drawn on the basis of the sample.

⊛ Variable: The measurements of elements of a population having certain characteristic may vary from element to element either in magnitude or in quality. These measurable characteristics are called variables. Two types:
1) Qualitative variable: ex- outcome of coin toss, sex of person.
2) Quantitative variable: ex- height of plant, yield of crops.
   a) Discrete variable: when the variable can only be integer. Number of plant.
   b) Continuous variable: Any values within a range. plant height.

- Attribute - The characteristic used to classify an individual into different categories.
- Classification is the process of arranging individuals in groups according to their affinities.
- Tabulation involves the orderly presentation of numerical facts in tabular form.

Scale of Measurement:

Scale of measurement is a system used to classify and measure data according to their property, order and numerical meaning. 4 types:

1) Nominal Scale:
This classify data into different categories or group without any particular order or ranking. It only tells us weather observation belong to the same or different categories. Example: crop type. rice, wheat.

2) Ordinal Scale:
classify data into categories that have a meaningful order or ranking but the difference between categories is not necessary equal. We can tell which category is higher or lower. Ex. Disease severity- Low, high, medium.

3) Interval scale:
Order difference with equal difference between two successive values. But it doesn't have a true zero point.

Difference between values are meaningful, but ratios are not meaningful because zero doesn't mean complete absence of the quantity.

Example: difference between 20°C and 30°C is the same as between 30°C and 40°C but 40°C is not twice as hot as 20°C.

4. Ratio scale:

A ratio scale has ordered value, equal interval and true zero point. Both difference and ratio are meaningful because zero represent absolute absence. Ex: Rainfall, height. 4m height is twice than 2m height of plant.

- Parameter - measurement of population
- statistics - measurement of sample.

Frequency distribution:

Most important form of tabulation.
This is a form of condensing and summarising a long and complex record of quantitative data.

Class interval: Instead of considering the frequency for each value of x, we can obtain the frequency for a certain interval of marks, eg $25 \le x \le 29$, $30 \le x \le 34$ and so on. These intervals of the variable x are known as the class interval of x.

- class frequency: No of observation in a particular class.
- The lower and upper value of the class is called lower limit and upper limit respectively. Difference between them is called length of class interval. The average of lower and upper limit of the class interval is called mid value.

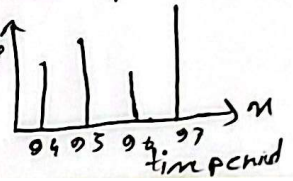
- Construction of a frequency distribution:
  1) Find out range by subtracting the lower value from the highest value of the variable x.
  2) Divide the range by 5 and 25, round these numbers. Call this number a and b. The class interval should normally between a and b.
  3) Arrange the sheet with three headings: class interval, tally marks and freq^n.
  4) Read the item one by one and put a tally mark against appropriate class-interval
  5) Count the number of tally marks opposite each interval and write the result in the frequency column.

H.A stages, Range, $R = X_{(n)} - X_{(1)}$ and class interval $= \frac{R}{K}$ where $K = 1 + 3.3 \log_{10} N$

N → Total no. of observation

Graphical Representation:

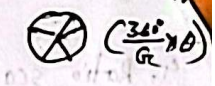
a) Line diagram: Usually time series data, where time is usually plotted in x-axis (horizontal) and the value of plotted to the y axis (vertical). The ploted points are added by a straight line to the x axis which is parallel to the y axis.



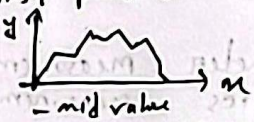
ⓑ Bar diagram: The qualitative characters are ploted in the x axis (H) and the value of different level are ploted in the y axis(V) with proper scale. The value of the character is shown by the rectangle instead of aline.



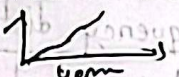
ⓒ Pie diagram: The value of different levels of a qualitative variable are presented in a circle containing the angle for each level separately. where the total angle is 360°.



ⓓ Frequency polygon: The mid value of the continuous class interval are represented along x-axis and the frequency corresponding to the class intervals are represented along the y-axis. Then the points are added by straight line. The first point and last point are joining to the mid value of next class interval.

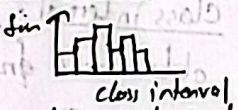


ⓔ Cumulative frequency polygon: The upper limit of the class interval are plotted in the x axis and the cumulative frequency at y axis, added by line (straight)



ⓕ Ogive: when the points are added by free hand to smooth the cumulative frequency polygon.

ⓖ Histogram: X→ class interval Y→ Frequency (as each class has a with so the frequency are shown with bars. Histogram is particularly appropriate for continuous data

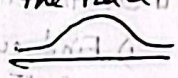


Frequency Curve:

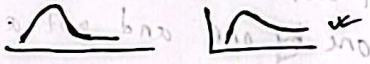
If the no of observation are large and length of the class interval can be reduced The frequency polygon will produce a smooth curve usually called frequency curve.

4 types:
a) Symetrical curve
b) Moderately asymmetrical or skew curve.
c) Extremely asymmetrical or J-shaped curve.
d) U-shaped curve.

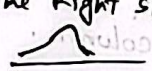
- Symetrical Curve: The frequency at the mid position is maximum and the rate of decrease from the peak is same in both side



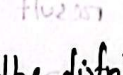
- Skew curve: Not symetric. The rate of decrease from the top is not same in the both side:
  - ✓ If the rate of decrease is rapid in the left side and a longer tail on the right is called positively skew curve.



  - ✓ If the rate of decrease is rapid in the Right side and longer tail on the left side is called negetively skew curve.



- J shape: If the maximum frequency occurs at the one end of the distribution.
  - ✓ If this occurs at the left end then it is positively J-shaped
  - ✓ " " " " " right " " " negetively J shaped



- U shape curve: The maximum frequency occurs at the bothend and the minimum in the middle. Distibution of the human death followed U shape curve

